



Database Management Systems

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Normalization and Decomposition

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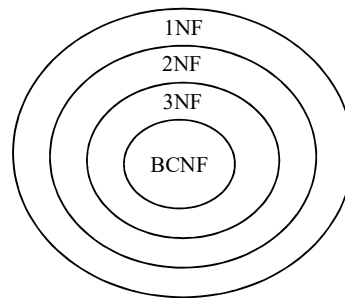
Normalization



❑ **Normalization** is a systematic approach to organize data in a database to eliminate redundancy, avoid anomalies and ensure data consistency.

• Normal Forms:

- 1NF
- 2NF
- 3NF
- BCNF



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1st Normal Form (1NF)



- ❑ **The initial step of normalization.**
- ❑ By default when you convert an E-R diagram to relations it will be in 1NF.
- ❑ The relation is said to be in 1NF if:
 - Each cell in a table contains atomic values.
 - Each column of a relation have unique names.
 - Every relation have a primary key.

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1st Normal Form (1NF)



Un-Normalized

S _{ID}	S _{Name}	C _{Name}
101	Vineet	DBMS, C++
102	Ayush	ALGO
103	Surya	Micro

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1st Normal Form (1NF)



Un-Normalized

S _{ID}	S _{Name}	C _{Name}
101	Vineet	DBMS, C++
102	Ayush	ALGO
103	Surya	Micro

1NF

S _{ID}	S _{Name}	C _{Name}
101	Vineet	DBMS
101	Vineet	C++
102	Ayush	ALGO
103	Surya	Micro

S _{ID}	S _{Name}	S _{ID}	C _{Name}
101	Vineet	101	DBMS
102	Ayush	101	C++
103	Surya	102	ALGO
		103	Micro

S _{ID}	S _{Name}	C _{Name1}	C _{Name2}
101	Vineet	DBMS	C++
102	Ayush	ALGO	Null
103	Surya	Micro	Null

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2nd Normal Form (2NF)



- ☐ The relation R is said to be in 2NF if:
 - Relation R is in 1NF.
 - R should not contain any partial dependency (every non-prime attributes should be fully dependent upon candidate key).
- ☐ In simple word “each column should be directly related to the primary key, and not to other columns.”
- ☐ How to achieve 2NF?
 - Identify composite key
 - Locate partial dependency
 - Decompose table/relation

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2nd Normal Form (2NF)



- ☐ **Super Keys:** A super key is a unique set of attribute(s) that determine the set of other attributes in a relation.
- ☐ Super Keys: {E_{ID}}, {S_{ID}, C_{ID}}, {S_{ID}, C_{ID}, S_{Name}}, {E_{ID}, S_{ID}}, etc.
- ☐ Invalid: {S_{ID}} {C_{ID}, C_{Name}}

E _{ID}	S _{ID}	S _{Name}	C _{ID}	C _{Name}
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

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2nd Normal Form (2NF)



- ❑ Super Keys: $\{E_{ID}\}$, $\{S_{ID}, C_{ID}\}$, $\{S_{ID}, C_{ID}, S_{Name}\}$, $\{E_{ID}, S_{ID}\}$, etc.
- ❑ **Candidate Key**: is a minimal super key (removing any attribute from a candidate key will not retain its ability to uniquely determine other attributes)

- ❑ Candidate Keys: $\{E_{ID}\}$, $\{S_{ID}, C_{ID}\}$

<u>E_{ID}</u>	<u>S_{ID}</u>	<u>S_{Name}</u>	<u>C_{ID}</u>	<u>C_{Name}</u>
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

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2nd Normal Form (2NF)



- ❑ Candidate Keys: $\{E_{ID}\}$, $\{S_{ID}, C_{ID}\}$.
- ❑ **Prime Attribute**: an attribute which is part of candidate key attribute.
- ❑ **Non-Prime Attribute**: an attribute which is not part of any of the candidate key.

- ❑ Prime Attributes: E_{ID}, S_{ID}, C_{ID}
- ❑ Non-Prime Attributes: S_{Name}, C_{Name}

<u>E_{ID}</u>	<u>S_{ID}</u>	<u>S_{Name}</u>	<u>C_{ID}</u>	<u>C_{Name}</u>
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

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2nd Normal Form (2NF)



- ❑ Candidate Keys: $\{E_{ID}\}$, $\{S_{ID}, C_{ID}\}$.
- ❑ Prime Attributes: E_{ID} , S_{ID} , C_{ID}
- ❑ Non-Prime Attributes: S_{Name} , C_{Name}

E_{ID}	S_{ID}	S_{Name}	C_{ID}	C_{Name}
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

- ❑ **Fully Functional Dependency:** a non-prime attribute depends on the entire candidate key, not just part of it.

- $E_{ID} \rightarrow S_{ID}$
- $E_{ID} \rightarrow C_{ID}$
- $E_{ID} \rightarrow C_{Name}$
- $E_{ID} \rightarrow S_{Name}$
- $\{S_{ID}, C_{ID}\} \rightarrow \{S_{Name}, C_{Name}\}$

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2nd Normal Form (2NF)



- ❑ Candidate Keys: $\{E_{ID}\}$, $\{S_{ID}, C_{ID}\}$.
- ❑ Prime Attributes: E_{ID} , S_{ID} , C_{ID}
- ❑ Non-Prime Attributes: S_{Name} , C_{Name}

E_{ID}	S_{ID}	S_{Name}	C_{ID}	C_{Name}
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

- ❑ **Partial Functional Dependency:** If the proper subset of the candidate key determines a non-prime attribute, it is called partial dependency.

- $S_{ID} \rightarrow S_{Name}$ { S_{Name} only depends on S_{ID} }
- $C_{ID} \rightarrow C_{Name}$ { C_{Name} only depends on C_{ID} }

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2nd Normal Form (2NF)

☐ **Remove Partial dependency:** Create separate table for partial dependencies, and retain original relation.

E _{ID}	S _{ID}	S _{Name}	C _{ID}	C _{Name}
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

C _{ID}	C _{Name}
CS101	DBMS
CS102	C++
CS103	Micro

S _{ID}	S _{Name}
101	Vineet
102	Ayush
103	Surya

E _{ID}	S _{ID}	C _{ID}
1	101	CS101
2	101	CS103
3	102	CS102
4	102	CS101
5	103	CS103
6	103	CS102

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2nd Normal Form (2NF)

☐ **Now Check all relations.**

E _{ID}	S _{ID}	S _{Name}	C _{ID}	C _{Name}
1	101	Vineet	CS101	DBMS
2	101	Vineet	CS103	Micro
3	102	Ayush	CS102	C++
4	102	Ayush	CS101	DBMS
5	103	Surya	CS103	Micro
6	103	Surya	CS102	C++

C _{ID}	C _{Name}
CS101	DBMS
CS102	C++
CS103	Micro

S _{ID}	S _{Name}
101	Vineet
102	Ayush
103	Surya

E _{ID}	S _{ID}	C _{ID}
1	101	CS101
2	101	CS103
3	102	CS102
4	102	CS101
5	103	CS103
6	103	CS102

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2nd Normal Form (2NF)



- ☐ IF $R(A, B, C, D)$
- $AB \rightarrow CD$
 - Candidate Key: $\{AB\}$
 - Prime Attribute: A, B
 - Non-Prime Attribute: C, D
- ☐ D is completely dependent on key?
- Yes, fully dependency.
- ☐ C is completely dependent on key?
- Yes, Full dependency.

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2nd Normal Form (2NF)



- ☐ IF $R(A, B, C, D)$
- $AB \rightarrow D, A \rightarrow C$
 - Candidate Key: $\{AB\}$
 - Prime Attribute: A, B
 - Non-Prime Attribute: C, D
- ☐ D is completely dependent on key?
- Yes, fully dependency.
- ☐ C is completely dependent on key?
- No, $\{C$ is dependent on subset of candidate key $AB\}$, partial dependency.

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2nd Normal Form (2NF)



□ *IF* $R(A, B, C)$

- $B \rightarrow C$
- Candidate Key $\rightarrow \{AB\}$
- Prime Attribute: A, B
- Non-Prime Attribute: C

□ C is completely dependent on key?

- No, {C is dependent on subset of candidate key AB}, partial dependency.

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2nd Normal Form (2NF)



□ *IF* $R(A, B, C, D)$

- $A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A$
- Candidate Key: {A} {B} {C}, {D}
- Prime Attribute: A, B, C, D
- Non-Prime Attribute: {Null}

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2nd Normal Form (2NF)



- *IF R(A, B, C, D, E)*
- $A \rightarrow B, BC \rightarrow D, F \rightarrow C, D \rightarrow A$
 - Candidate Key: {AE} {DE} {BE}
 - Prime Attribute: A, B, D, E
 - Non-Prime Attribute: C

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2nd Normal Form (2NF)



- *IF R(A, B, C, C)*
- $A \rightarrow B, B \rightarrow C, C \rightarrow D$
 - Candidate Key $\rightarrow \{A\}$
 - Prime Attribute: A
 - Non-Prime Attribute: B, C, D
- C is completely dependent on key?
- No, {C is dependent on subset of candidate key AB}, partial dependency.

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2nd Normal Form (2NF)

☐ **IF R(A, B, C)**

- $B \rightarrow C$
- Composite Key $\rightarrow AB$
- Prime Attribute: A, B
- Non-Prime Attribute: C

A	B	C
a	1	x
b	2	y
a	3	z
c	3	z
d	3	z
e	3	z

A	B
a	1
b	2
a	3
c	3
d	3
e	3

B	C
1	x
2	y
3	z

☐ C is completely dependent on key?

- No, {C is dependent on subset of candidate key AB}, partial dependency.

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2nd Normal Form (2NF)

☐ **IF R(A, B, C, D)**

- $A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A$
- Composite Key $\rightarrow AB$
- Prime Attribute: A, B
- Non-Prime Attribute: C

☐ C is completely dependent on key?

- No, {C is dependent on subset of candidate key AB}, partial dependency.

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2nd Normal Form (2NF)



☐ $R(A, B, C, D, E, F, G, H)$

- $A \rightarrow B, C \rightarrow DE, E \rightarrow FG, G \rightarrow H, AC \rightarrow ABCDEFG$

☐ Candidate Keys:

- AC

☐ Non-Prime Attributes:

- B, D, E, F, G, H

☐ Partial Dependency:

- $A \rightarrow B, C \rightarrow DE$

☐ Decomposition

- One table for $A \rightarrow B$,
- Another table for $C \rightarrow DE$
- And the main table.

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Normalization Example: 1



- Is the given relation is in 1NF?
- Is the given relation is in 2NF?

S _{ID}	S _{Name}	C _{ID}	C _{Name}	Prof.
101	Vineet	CS101	DBMS	DNY
101	Vineet	CS102	Algo	SP
102	Ayush	CS101	DBMS	DNY
102	Ayush	CS103	Python	PD

- Candidate Key: (S_{ID}, C_{ID})
- Functional Dependencies: S_{ID} → S_{Name}, C_{ID} → C_{Name}, Prof.
- Prime Attribute: S_{ID}, C_{ID}
- Non-Prime Attribute: S_{Name}, C_{Name}, Prof.
- Is there any partial dependencies: No

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Normalization Example: 1



- Is the table is redundant?

S _{ID}	S _{Name}	C _{ID}	C _{Name}	Prof.
101	Vineet	CS101	DBMS	DNY
101	Vineet	CS102	Algo	SP
102	Ayush	CS101	DBMS	DNY
102	Ayush	CS103	Python	PD

- Why it can be problematic if the table is redundant: Anamolies.
- Type of Anamolies: Insertion, Deletion and Update Anamolies.
- **Insertion: Cannot insert a course unless at least one student is enrolled.**
- **Updation: Any change in course name, instructor, must be reflected in all rows.**
- **Deletion: Deletion a record (last one) delete the course detail.**

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Example 2



□ *IF R(A, B, C)*

- $B \rightarrow C$
- Is the table is in 1NF?
- Is the table is in 2NF?
- Candidate Key: AB
- Prime Attribute: A, B
- Non-Prime Attribute: C
- Is there any partial dependencies exist?

A	B	C
a	1	x
b	2	y
a	3	z
c	3	z
d	3	z
e	3	z

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Example 2



□ *IF R(A, B, C)*

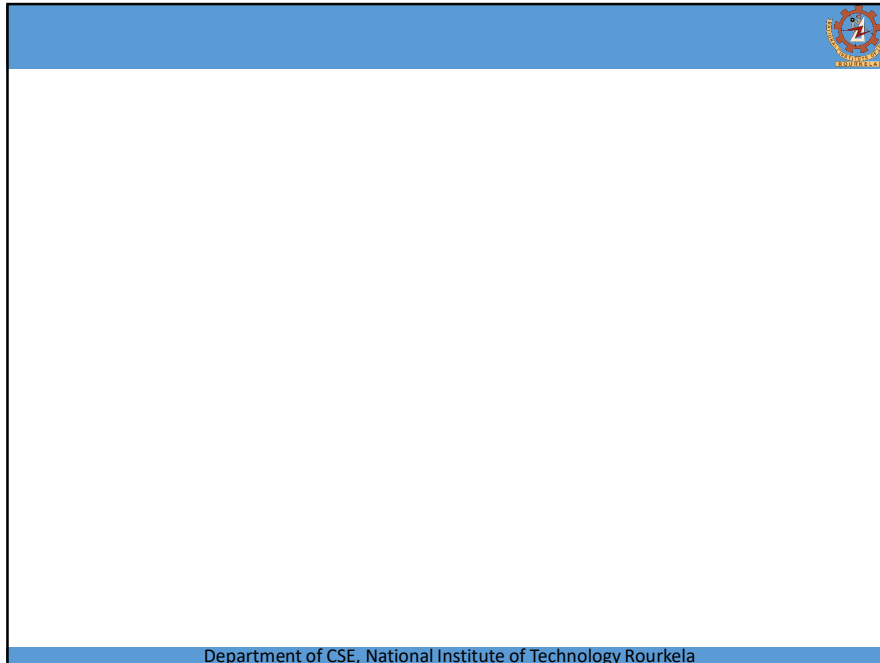
- $B \rightarrow C$
- How we can Normalize the table into 2NF.

A	B	C
a	1	x
b	2	y
a	3	z
c	3	z
d	3	z
e	3	z

A	B
a	1
b	2
a	3
c	3
d	3
e	3

B	C
1	x
2	y
3	z

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3rd Normal Form (3NF)

- ☐ The relation R is said to be in 3NF if:
 - Relation R is in 2NF.
 - There should be **no transitive dependency in R.**
- ☐ A relational schema is said to be in 3NF if every functional dependency $A \rightarrow B$ in R; either A is **super key or candidate key** or B is the **prime attribute**.
- ☐ **Transitive Dependency:** A functional dependency from non-prime attribute to non-prime attribute is called transitive functional dependency.
- ☐ How to achieve 3NF?
 - Identify transitive dependency
 - Decompose table/relation

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3rd Normal Form (3NF)



☐ R (S_{ID}, State, City), FD: {S_{ID}→State, State→City}

☐ Candidate Key: S_{ID}

☐ Prime Attribute: S_{ID}

☐ Non-Prime Attributes: State, City

☐ Partial Dependency: No

☐ Transitive Dependency: State→City

S _{ID}	State	City
101	OD	BBS
102	HR	Ambala
103	OD	Bhadrak
104	CG	Durg
105	CG	Raipur
106	OD	BBS
107	CG	Raipur

☐ Prime → Non Prime {No Problem}

☐ Non-Prime→ Prime {No Problem}

☐ Non-Prime → Non-Prime {Transitive Dependency}

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3rd Normal Form (3NF)



☐ Remove transitive dependency:

S _{ID}	State	City
101	OD	BBS
102	HR	Ambala
103	OD	Bhadrak
104	CG	Durg
105	CG	Raipur
106	OD	BBS
107	CG	Raipur

S _{ID}	City
101	BBS
102	Ambala
103	Bhadrak
104	Durg
105	Raipur
106	BBS
107	Raipur

State	City
OD	BBS
HR	Ambala
OD	Bhadrak
CG	Durg
CG	Raipur

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Normalization Example: 1



- What are the transitive dependency in this table?
- $S_{ID} \rightarrow S_{Name}$: Why? S_{ID} is part of composite key, but not the both attributes?
- $C_{ID} \rightarrow C_{Name}$, Prof. : Same as above.

S_{ID}	S_{Name}	C_{ID}	C_{Name}	Prof.
101	Vineet	CS101	DBMS	DNY
101	Vineet	CS102	Algo	SP
102	Ayush	CS101	DBMS	DNY
102	Ayush	CS103	Python	PD

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Normalization Example: 1



- How it will be converted to 3NF?

S_{ID}	S_{Name}	C_{ID}	C_{Name}	Prof.
101	Vineet	CS101	DBMS	DNY
101	Vineet	CS102	Algo	SP
102	Ayush	CS101	DBMS	DNY
102	Ayush	CS103	Python	PD

S_{ID}	S_{Name}
101	Vineet
101	Vineet
102	Ayush
102	Ayush

C_{ID}	C_{Name}	Prof.
CS101	DBMS	DNY
CS102	Algo	SP
CS101	DBMS	DNY
CS103	Python	PD

S_{ID}	C_{ID}
101	CS101
101	CS102
102	CS101
102	CS103

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Normalization Example: 1



- $R(A, B, C, D)$
- $AB \rightarrow CD, D \rightarrow A$

- What are the candidate keys: AB, BD
- Prime Attribute: A, B, D
- Non-Prime Attribute: C
- Is there any partial dependencies: No
- Transitive Dependency: No

- Remember the rule:

- If LHS of all FDs are Either Super Key or Candidate Key
- OR RHS is a Prime Attribute.

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Boyce-Codd Normal Form (BCNF)



- ☐ 3NF: LHS of all FDs are *Either Super Key or Candidate Key* OR RHS is a *Prime Attribute*.
- ☐ Special case of 3NF
- ☐ The relation R is said to be in BCNF if:
 - Relation R is in 3NF.
 - There should be no transitive dependency in R.
- ☐ For every functional dependency $X \rightarrow Y$, X must be a super key/candidate key (**the or part is removed**).

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BCNF



- ☐ Candidate Key: $\{S_{ID}\}, \{V_{ID}\}$
- ☐ FD = $\{S_{ID} \rightarrow S_{Name}, S_{ID} \rightarrow V_{ID}, V_{ID} \rightarrow age, V_{ID} \rightarrow S_{ID}\}$
- ☐ Check for the condition: For each FD, LHS should be Super Key or Candidate Key.

S_{ID}	S_{Name}	V_{ID}	Age
101	Ravi	V01	20
102	Varun	V02	19
103	Ravi	V03	21
104	Rahul	V04	22

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BCNF Example



- R (A, B, C, D)
- $AB \rightarrow CD, D \rightarrow A$
- R(ABCD) will divided into R1(D,A), R2(B, C, D)

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BCNF Example 2



- $R(A, B, C)$
- $AB \rightarrow C, C \rightarrow B$
- What are the candidate keys: AB, ~~BC~~, **AC**
- If LHS of all FDs are Either Super Key or Candidate Key
- $R(A, B, C) \rightarrow R_1(A, B); R_2(C, B)$ **AC CB**

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Question



- Q1. $R(A, B, C, D, E, F, G, H, I, J)$; FD $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$
- Keys?
 - Decompose it to 2NF

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Decomposition Rule



- ☐ Decomposition should be lossless: (No data loss: fulfilled by 1NF, 2NF, 3NF, BCNF)
- ☐ Dependency should be preserved (Sometime not fulfilled by BCNF).

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Decomposition Example



- R (A, B, C, D)
- $AB \rightarrow CD, D \rightarrow A$
- R(ABCD) will divided into R1(D,A), R2(B, C, D)
- In R1 we have $D \rightarrow A$, but what about $AB \rightarrow CD$? It is not available in both relations.

AB BD

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Questions



- ☐ When we can conclude that the relation is in 1NF?
- ☐ How to convert a relation into 1NF?
- ☐ When we can conclude that any relation is in 2NF?
- ☐ What is partial dependency?
- ☐ How to convert a relation into 2NF?
- ☐ When we can conclude that the relation is in 3NF?
- ☐ What is transitive dependency?
- ☐ When to say that the relation is in BCNF?
- ☐ What is non-trivial dependency? $X \rightarrow Y$ $Y \not\rightarrow X$
- ☐ Consider a relation $R(A, B, C, D)$, the primary key for the relation is $\{AB\}$, is that possible that value of A or B can be null?

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Questions



- ☐ The first reason we have already discussed (we will have many redundant data which can cause Anamolies)
- ☐ It is possible that the dependency given for the relation may not hold for some instances.
- ☐ Consider a relation $R(A, B, C, D)$ and $FD = \{AB \rightarrow D, B \rightarrow C\}$
- ☐ The candidate key for this relation is AB, the same will be primary key.
- ☐ The only condition for primary key is it should be unique.
- ☐ Thus the value of either A or B can be null, If the value of B is null the dependency $B \rightarrow C$ will be void.

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Questions



- ☐ Consider a relation $R(A, B, C, D)$ and $FD = \{AB \rightarrow C, C \rightarrow D\}$
- ☐ What will be candidate key and primary key for the relation?
- ☐ Is the relation is in 2NF or not?
- ☐ Is the relation is in 3NF?
- ☐ What about BCNF?
- ☐ What is the problem if the relation is not in 3NF?
 - Again for the first dependency $AB \rightarrow C$; AB combined is a primary key at least one of them will be not null; thus for sure this dependency will hold.
 - But what about $C \rightarrow D$; C may become null, and thus AB may not reach to D .
 - $R_1(A, B, C)$ & $R_2(C, D)$

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Questions



- ☐ Consider a relation $R(A, B, C, D, E)$ and $FD = \{A \rightarrow C, B \rightarrow E, C \rightarrow D\}$
- ☐ What will be candidate key and primary key for the relation?
- ☐ Is the relation is in 2NF or not?
- ☐ Is the relation is in 3NF?
- ☐ What about BCNF?
- ☐ How many relations we will get after converting this relation to 3NF?

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Questions



- ☐ Consider a relation $R(A, B, C)$ and $FD = \{AB \rightarrow C, C \rightarrow B\}$
- ☐ What will be candidate key and primary key for the relation?
- ☐ Is the relation is in 2NF?
- ☐ Is the relation is in 3NF?
- ☐ Is the relation is in BCNF?
- ☐ Relations after decomposition?

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Questions



- ☐ If you are asked to identify NF of a relation then from which NF (1NF, 2NF, 3NF, BCNF) will you start and why?
- ☐ Identify normal forms for this relations.
- ☐ $R(ABCEFGH)$, $FD = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH\}$
 - 1NF
- ☐ $R(ABCDEF)$, $FD = \{AB \rightarrow C, DC \rightarrow AE, E \rightarrow F\}$
 - 1NF
- ☐ $R(ABCDEF)$, $FD = \{AB \rightarrow C, DC \rightarrow AE, E \rightarrow F\}$
 - 1NF
- ☐ $R(ABCDEFGHI)$, $FD = \{AB \rightarrow C, BD \rightarrow EF, AD \rightarrow GH, A \rightarrow I\}$
 - 1NF
- ☐ $R(ABCDE)$, $FD = \{AB \rightarrow CD, D \rightarrow A, BC \rightarrow DE\}$
 - 3NF

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Questions



- ☐ Identify normal forms for this relations.
- ☐ $R(ABCDE)$, $FD = \{BC \rightarrow ADE, D \rightarrow B\}$
 - 3NF
- ☐ $R(VWXYZ)$, $FD = \{X \rightarrow YV, Y \rightarrow Z, Z \rightarrow Y, VW \rightarrow X\}$
 - 1NF
- ☐ $R(ABCDEF)$, $FD = \{ABC \rightarrow D, ABD \rightarrow E, CD \rightarrow F, CDF \rightarrow B, BF \rightarrow D\}$
 - 1NF
- ☐ $R(ABC)$, $FD = \{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$ [GATE]
 - BCNF

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GATE Questions



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GATE CSE 2013



- ☐ R (ABCDEFGH), FD = {CH→G, A→BC, B→CFH, E→A, F→EG}
- ☐ The relation is in which normal form

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GATE CSE 2024



- ☐ Which of the following is true if the relation is in 1NF?
 - ☐ R can have multi-attribute key.
 - ☐ R cannot have a foreign key.
 - ☐ R cannot have composite attribute.
 - ☐ R cannot have more than one candidate key.

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GATE CSE 2012

4) Which of the following is TRUE?

- (A) Every relation in 2NF is also in BCNF
- (B) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
- (C) Every relation in BCNF is also in 3NF
- (D) No relation can be in both BCNF and 3NF

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GATE CSE 2008

4) Consider the following relational schemes for a library database:

Book (Title, Author, Catalog_no, Publisher, Year, Price)

Collection (Title, Author, Catalog_no)

with in the following functional dependencies:

- I. Title Author --> Catalog_no
- II. Catalog_no --> Title Author Publisher Year
- III. Publisher Title Year --> Price

Assume {Author, Title} is the key for both schemes. Which of the following statements is true?

- (A) Both Book and Collection are in BCNF
- (B) Both Book and Collection are in 3NF only
- (C) Book is in 2NF and Collection is in 3NF
- (D) Both Book and Collection are in 2NF only

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GATE CSE 2008

1) Which one of the following statements about normal forms is FALSE?

- (a) BCNF is stricter than 3NF
- (b) Lossless, dependency-preserving decomposition into 3NF is always possible
- (c) Lossless, dependency-preserving decomposition into BCNF is always possible
- (d) Any relation with two attributes is in BCNF

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For the relation $R(ABCDEFGH)$ with FD's=
 $\{CH \rightarrow G, A \rightarrow BC, B \rightarrow CHF, E \rightarrow A, F \rightarrow EG\}$ such
that F^+ is exactly the set of FDs that hold for
 R .} Consider the FDs given in above question.
The relation R is

- ☐ in 1NF, but not in 2NF.
- ☐ in 2NF, but not in 3NF.
- ☐ in 3NF, but not in BCNF.
- ☐ in BCNF

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Question 2

Which of the following is **TRUE**?

☐ Every relation in 3NF is also in BCNF

☐ A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R

☐ Every relation in BCNF is also in 3NF

☐ No relation can be in both BCNF and 3NF

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Consider a relational table with a single record for each registered student with the following attributes.

1. *Registration_Num*: Unique registration number of each registered student
2. *UID*: Unique identity number, unique at the national level for each citizen
3. *BankAccount_Num*: Unique account number at the bank. A student can have multiple accounts or join accounts. This attribute stores the primary account number.
4. *Name*: Name of the student
5. *Hostel_Room*: Room number of the hostel

Which one of the following option is **INCORRECT**?

☐ BankAccount_Num is candidate key

☐ Registration_Num can be a primary key

☐ UID is candidate key if all students are from the same country

☐ If S is a superkey such that S ∩ UID is NULL then S ∪ UID is also a superkey

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Consider the following relational schema:

Suppliers(sid:integer, sname:string, city:string, street:string)
Parts(pid:integer, pname:string, color:string)
catalog(sid:integer, pid:integer, cost:real)

Assume that, in the suppliers relation above, each supplier and each street within a city has a unique name, and (sname, city) forms a candidate key. No other functional dependencies are implied other than those implied by primary and candidate keys. Which one of the following is TRUE about the above schema?

☒ The schema is in BCNF

☐ The schema is in 3NF but not in BCNF

☐ The schema is in 2NF but not in 3NF

☐ The schema is not in 2NF

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Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies $\{\{E, F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, K \rightarrow \{M\}, L \rightarrow \{N\}\}$ on R . What is the key for R ?

☐ $\{E, F\}$

☐ $\{E, F, H\}$

☐ $\{E, F, H, K, L\}$

☐ $\{E\}$

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Given the following two statements:

S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF and BCNF.

S2: $AB \rightarrow C$, $D \rightarrow E$, $E \rightarrow C$ is a minimal cover for the set of functional dependencies $AB \rightarrow C$, $D \rightarrow E$, $AB \rightarrow E$, $E \rightarrow C$.

Which one of the following is CORRECT?

☐ S1 is TRUE and S2 is FALSE.

☐ Both S1 and S2 are TRUE.

☐ S1 is FALSE and S2 is TRUE.

☐ Both S1 and S2 are FALSE.



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Question 9

Given the STUDENTS relation as shown below.

<i>StudentID</i>	<i>StudentName</i>	<i>StudentEmail</i>	<i>StudentAge</i>	<i>CPI</i>
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

For (StudentName, StudentAge) to be the key for this instance, the value X should not be equal to



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The relation scheme Student Performance (name, courseNo, rollNo, grade) has the following functional dependencies:

```

name, courseNo → grade
rollNo, courseNo → grade
name → rollNo
rollNo → name

```

The highest normal form of this relation scheme is

☐ 2 NF
☒ 3 NF
☐ BCNF
☐ 4NF

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Consider the following functional dependencies in a database:

```

Data_of_Birth → Age
Age → Eligibility
Name → Roll_number
Roll_number → Name
Course_number → Course_name
Course_number → Instructor
(Roll_number, Course_number) → Grade

```

The relation (Roll_number, Name, Date_of_birth, Age) is:

☐ In second normal form but not in third normal form
☐ In third normal form but not in BCNF
☐ In BCNF
☐ None of the above

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Relation R is decomposed using a set of functional dependencies, F and relation S is decomposed using another set of functional dependencies G. One decomposition is definitely BCNF, the other is definitely 3NF, but it is not known which is which. To make a guaranteed identification, which one of the following tests should be used on the decompositions? (Assume that the closures of F and G are available).

- ☐ Dependency-preservation
- ☐ Lossless-join
- ☐ BCNF definition
- ☐ 3NF definition

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Consider a schema $R(A,B,C,D)$ and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Then the decomposition of R into $R_1(AB)$ and $R_2(CD)$ is

- ☐ dependency preserving and lossless join
- ☐ lossless join but not dependency preserving
- ☐ dependency preserving but not lossless join
- ☐ not dependency preserving and not lossless join

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Decomposition

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Decomposition

- ❑ **Normalization:** decomposition of tables in smaller table.
- ❑ **Rule-1 (Mandatory):** Decomposition should be Lossless.
 - **Lossless:** if we combine the child table we must get parent table.
 - **Lossy:** if we do not get the parent table (actual tuple) then lossy.
 - Problem with lossy decomposition: (a) we may not get original tuple.
(b) We may get redundant tuple.
- ❑ **Rule 2 (Optional):** Decomposition should preserve original dependency.

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Lossless Decomposition



- If a relation R is decomposed into two relations R1 and R2, then the decomposition will be **lossless iff**:
- $\text{Attribute}(R1) \cup \text{Attribute}(R2) = \text{Attribute}(R) \rightarrow$ An attribute must be present in at least one table.
 - $\text{Attribute}(R1) \cap \text{Attribute}(R2) \neq \emptyset \rightarrow$ There must be a common attribute between the decomposed relations.
 - $\text{Attribute}(R1) \cap \text{Attribute}(R2) \rightarrow \text{Attribute}(R1) \text{ or } \text{Attribute}(R2) \rightarrow$ the common attribute must be a **candidate key in one of the decomposed relation**.

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Decomposition



- **Condition (a):** $\text{Attribute}(R1) \cup \text{Attribute}(R2) = \text{Attribute}(R) \rightarrow$ An attribute must be present in at least one table.
- We will only get the attributes A B and C if we combine the decomposed relation.

A	B	C	D
1	2	1	X
2	2	2	Y
3	3	2	Z

A	B
1	2
2	2
3	3

C
1
2
2

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Decomposition



- ❑ **Condition (b):** $\text{Attribute}(R1) \cap \text{Attribute}(R2) \neq \emptyset \rightarrow$ There must be at least one common attribute between the decomposed relations.
- ❑ **Query:** find the value of C if the value of a is 1.
- ❑ We cannot write an equivalent query for this as there is no common attribute for checking.

A	B	C	D
1	2	1	X
2	2	2	Y
3	3	2	Z

A	B
1	2
2	2
3	3

C	D
1	X
2	Y
2	Z

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Decomposition



- ❑ **Condition (b):** $\text{Attribute}(R1) \cap \text{Attribute}(R2) \neq \emptyset \rightarrow$ There must be at least one common attribute between the decomposed relations.
 - ❑ **Query:** find the value of C if the value of a is 1.
- Select R2.C from R2 Natural Join Where R1.A = 1.

A	B	B	C
1	2	2	1
1	2	2	2
1	2	3	2
2	2	2	1
2	2	2	2
2	2	3	2
3	3	2	1
3	3	2	2
3	3	3	2

A	B	C
1	2	1
1	2	2
2	2	1
2	2	2
3	3	2

A	B	C
1	2	1
2	2	2
3	3	2

A	B
1	2
2	2
3	3

B	C
2	1
2	2
3	2

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Decomposition

❑ **Condition (c):** $\text{Attribute}(R1) \cap \text{Attribute}(R2) \rightarrow \text{Attribute}(R1)$ or $\text{Attribute}(R1) \cap \text{Attribute}(R2) \rightarrow \text{Attribute}(R2)$ → the common attribute must be a candidate key in one of the decomposed relation.

❑ **Query:** find the value of C if the value of a is 1.

Select R2.C from R2 Natural Join Where R1.A = 1.

A	B	A	C
1	2	1	1
1	2	2	2
1	2	3	2
2	2	1	1
2	2	2	2
2	2	3	2
3	3	1	1
3	3	2	2
3	3	3	2

A	B	C
1	2	1
2	2	2
3	3	2

A	B	C
1	2	1
2	2	2
3	3	2

A	B
1	2
2	2
3	3

A	C
1	1
2	2
3	2

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Lossless Decomposition: Example 1

❑ For the given relation check which of the following decomposition is lossless.

- R1(AB), R2(CD)
- R1(ABC), R2(DE)
- R1(ABC), R2(CDE)
- R1(ABCD), R2(ACDE)
- R1(ABCD), R2(DE)
- R1(ABC), R2(BCD), R3(DE)

A	B	C	D	E
a	123	1	p	w
b	234	2	q	x
a	345	1	r	y
c	456	3	s	z

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Lossless Decomposition: Example 2



□ $R(VWXYZ), FD = \{Z \rightarrow Y, Y \rightarrow Z, X \rightarrow YV, VW \rightarrow X\}$.

- $R1(VWX), R2(XYZ)$
- $R1(VW), R2(YZ)$
- $R1(VWX), R2(YZ)$
- $R1(VW), R2(WXYZ)$

$X \rightarrow YZ$

A	B	C	D	E
a	123	1	p	w
b	234	2	q	x
a	345	1	r	y
c	456	3	s	z

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Dependency Preserving Decomposition



□ If a relation R , having FD set F , is decomposed into two relations $R1$ and $R2$ having FD set $F1$ and $F2$ then the decomposition is said to be dependency preserving decomposition iff:

$F1, F2 \subseteq F^+$ and $(F1 \cup F2)^+ = F^+$

□ $R(ABC) F = \{A \rightarrow B, B \rightarrow C, C \rightarrow A\}$

- $R1(A, B)$ $R2(B, C)$
- $F1(A \rightarrow B, B \rightarrow A)$ $F2(B \rightarrow C, C \rightarrow B)$

• $(F1 \cup F2)^+ \rightarrow A \rightarrow B, B \rightarrow C, C \rightarrow A.$

□ Check the dependency based on the dependency given in parent table.

□ Do not put trivial dependency.

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Dependency Preservation [GATE]



- $R(ABCD) F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B\}$
- $R1(A,B); R2(B,C); R3(B,D)$
 - $R1(A,B);$ All possibilities $\{A \rightarrow A, A \rightarrow B, B \rightarrow B, B \rightarrow A\}$ $F1 \{A \rightarrow B\}$
 - $R2(B,C); F2 \{B \rightarrow C, C \rightarrow B\}$
 - $R3(B,D); F3 \{B \rightarrow D, D \rightarrow B\}$
 - $(F1 \cup F2) \rightarrow \{A \rightarrow B, B \rightarrow C, C \rightarrow B, B \rightarrow D, D \rightarrow B\}$
 - $(F1 \cup F2)^+ \rightarrow \{A \rightarrow B, B \rightarrow C, C \rightarrow D, B \rightarrow D, D \rightarrow B\}$

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Dependency Preservation



- $R(ABC) F = \{AB \rightarrow CD, D \rightarrow A\}$
- $R1(A, D)$ $R2(B,C,D)$
 - $F1(A \rightarrow D)$ $F2(BD \rightarrow C)$
 - $(F1 \cup F2)^+ \rightarrow D \rightarrow A$

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Decomposition to Normal Form



- $R(A, B, C, D, E, F, G, H, I, J)$
- $FD = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, D \rightarrow IJ, F \rightarrow GH\}$
 - Candidate Key?
 - AB
 - Normalize it to 2NF
 - $R_1(ABC);$
 - $R_2(ADEIJ);$
 - $R_3(BFGH)$

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Decomposition to Normal Form



- $R(A, B, C, D, E, F, G, H, I, J)$
- $FD = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, D \rightarrow IJ, F \rightarrow GH\}$
 - $R_1(ABC);$
 - $R_2(ADEIJ);$
 - $R_3(BFGH)$
 - Now decompose these relations to 3NF
 - $R_1(ABC);$
 - $R_{21}(ADE);$
 - $R_{22}(DIJ);$
 - $R_{31}(BF);$
 - $R_{32}(FGH)$
 - Now decompose these relations to BCNF.

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4 NF



☐ The relation should be in BCNF and it should not have multi-valued dependency.

A	B	C	D
1	X	P	2
1	Y	Q	3
1	X	P	3

☐ Candidate key → ABD,

☐ Which Normal Form → 3NF.

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4 NF



- ☐ Now check for BCNF
- ☐ Candidate key $\rightarrow$ ABD
- ☐ Dependency $B \rightarrow C$ Note allowed.
- ☐ Decompose it to two tables R1 (ABD), R2 (BC)
- ☐ Now the relation is in BCNF.

A	B	C	D
1	X	P	2
1	Y	Q	3
1	X	P	3

A	B	D
1	X	2
1	Y	3
1	X	2

B	C
X	P
Y	Q

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4 NF



- ☐ Multivalued Dependency ($\twoheadrightarrow$)
- ☐ $A \twoheadrightarrow B$
- ☐ $A \twoheadrightarrow D$
- ☐ Decompose the first relation to two more relations R1'(AB), R1''(AD)

A	B	C	D
1	X	P	2
1	Y	Q	3
1	X	P	3

A	B	D
1	X	2
1	Y	3
1	X	3

A	B
1	X
1	Y

A	D
1	2
1	3

B	C
X	P
Y	Q

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5 NF



- ☐ The relation should be in 4NF, and the decomposition should be lossless.
- ☐ Also known as Project-Join Normal Form (PJNF)
- ☐ We already discussed this concept yesterday.

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Normalization Summary



- ☐ Summary of Normal Forms:

NF	What It Fixes	Key Condition
1NF	Eliminates duplicate columns and repeating groups	Every attribute contains atomic (indivisible) values
2NF	Removes partial dependencies	Every non-key attribute is fully dependent on the entire primary key
3NF	Eliminates transitive dependencies	Every non-key attribute is only dependent on the primary key (not on other non-key attributes)
BCNF	Ensures every determinant is a superkey	Every functional dependency (FD) is on a superkey
4NF	Eliminates multi-valued dependencies (MVDs)	A table should not contain two independent multi-valued facts
5NF	Eliminates join dependencies (JDs)	A relation should not have redundancy due to join dependencies

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GATE Questions [2015]



Q5: Consider a relation $R(A,B,C)$ with a multivalued dependency $A \twoheadrightarrow B$. Which of the following decompositions is in 4NF?

- a. Decompose into $R_1(A,B)$ and $R_2(A,C)$
- b. Decompose into $R_1(A,B,C)$
- c. Decompose into $R_1(A,B)$ and $R_2(B,C)$
- d. Decompose into $R_1(A,C)$ and $R_2(B,C)$

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GATE Questions [2015]



Q5: Consider a relation $R(A,B,C)$ with a multivalued dependency $A \twoheadrightarrow B$. Which of the following decompositions is in 4NF?

- a. **Decompose into $R_1(A,B)$ and $R_2(A,C)$**
- b. Decompose into $R_1(A,B,C)$
- c. Decompose into $R_1(A,B)$ and $R_2(B,C)$
- d. Decompose into $R_1(A,C)$ and $R_2(B,C)$

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GATE Questions [2015]



Given a relation R with attributes A,B and C, and a multivalued dependency $A \twoheadrightarrow B$ is the decomposition of R into R1(A,B) and R2(A,C) in 4NF?

- a. Yes, because the decomposition removes the multivalued dependency.
- b. No, because the decomposition is not dependency-preserving.
- c. Yes, because $A \twoheadrightarrow B$ is preserved.
- d. No, because $A \twoheadrightarrow C$ is not preserved.

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GATE Questions [2015]



Given a relation R with attributes A,B and C, and a multivalued dependency $A \twoheadrightarrow B$ is the decomposition of R into R1(A,B) and R2(A,C) in 4NF?

- a. **Yes, because the decomposition removes the multivalued dependency.**
- b. No, because the decomposition is not dependency-preserving.
- c. Yes, because $A \twoheadrightarrow B$ is preserved.
- d. No, because $A \twoheadrightarrow C$ is not preserved.

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GATE Questions [2014]



Q1: Consider a relation $R(A,B,C,D)$ with functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Which of the following decompositions is both lossless and dependency-preserving?

- a. Decompose into $R_1(A,B)$ and $R_2(C,D)$
- b. Decompose into $R_1(A,C)$ and $R_2(A,D)$
- c. Decompose into $R_1(A,B,C)$ and $R_2(C,D)$
- d. Decompose into $R_1(A,B)$ and $R_2(A,C,D)$

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GATE Questions [2014]



Q1: Consider a relation $R(A,B,C,D)$ with functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Which of the following decompositions is both lossless and dependency-preserving?

- a. Decompose into $R_1(A,B)$ and $R_2(C,D)$ → lossy, DL
- b. Decompose into $R_1(A,C)$ and $R_2(A,D)$ → lossy, DL
- c. Decompose into $R_1(A,B,C)$ and $R_2(C,D)$ → lossless, DP
- d. Decompose into $R_1(A,B)$ and $R_2(A,C,D)$ → lossless, DL

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GATE Questions [2012]



Q2: Given a relation R with attributes A,B,C and D, and functional dependencies $A \rightarrow B$ and $C \rightarrow D$, is the decomposition of R into R1(A,B) and R2(C,D) lossless?

- a. Yes, because $A \rightarrow B$ and $C \rightarrow D$ are preserved.
- b. No, because there is no common attribute between R1 and R2
- c. Yes, because the decomposition is dependency-preserving.
- d. No, because A and C are not keys.

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GATE Questions [2012]



Q2: Given a relation R with attributes A,B,C and D, and functional dependencies $A \rightarrow B$ and $C \rightarrow D$, is the decomposition of R into R1(A,B) and R2(C,D) lossless?

- a. Yes, because $A \rightarrow B$ and $C \rightarrow D$ are preserved.
- b. No, because there is no common attribute between R1 and R2**
- c. Yes, because the decomposition is dependency-preserving.
- d. No, because A and C are not keys.

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GATE Questions [2011]



Q3: For a relation $R(A,B,C)$ with functional dependencies $AB \rightarrow C$ and $C \rightarrow A$ determine if the decomposition of R into $R_1(A,B)$ and $R_2(B,C)$ is lossless.

- a. Yes, because $AB \rightarrow C$ ensures lossless join.
- b. No, because $C \rightarrow A$ is not preserved.
- c. Yes, because $C \rightarrow A$ ensures lossless join.
- d. No, because $AB \rightarrow C$ is not preserved.



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GATE Questions [2011]



Q3: For a relation $R(A,B,C)$ with functional dependencies $AB \rightarrow C$ and $C \rightarrow A$ determine if the decomposition of R into $R_1(A,B)$ and $R_2(B,C)$ is lossless.

- a. Yes, because $AB \rightarrow C$ ensures lossless join.
- b. No, because $C \rightarrow A$ is not preserved.**
- c. Yes, because $C \rightarrow A$ ensures lossless join.
- d. No, because $AB \rightarrow C$ is not preserved.

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GATE Questions [2010]



Q4: Consider a schema $R(A,B,C,D)$ with functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Is the decomposition of R into $R_1(A,B)$ and $R_2(C,D)$ lossless?

- a. Yes, because each functional dependency involves a separate set of attributes.
- b. No, because there is no common attribute between R_1 and R_2 .
- c. Yes, because the decomposition is in BCNF.
- d. No, because the decomposition is not dependency-preserving.

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GATE Questions [2010]



Q4: Consider a schema $R(A,B,C,D)$ with functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Is the decomposition of R into $R_1(A,B)$ and $R_2(C,D)$ lossless?

- a. Yes, because each functional dependency involves a separate set of attributes.
- b. No, because there is no common attribute between R_1 and R_2 .**
- c. Yes, because the decomposition is in BCNF.
- d. No, because the decomposition is not dependency-preserving.

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